

END SEMESTER EXAMINATION 2025

n (CSE)

Name of the Paper: System Software

Time: 3 Hours

Semester: 5th
Paper Code: TCS501
MM: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

q1	Attempt Any two	20 Marks
(a)	The statement "Assemblers are machine-dependent programs" is often made in the context of system software. Do you agree with this statement? Justify your answer with appropriate reasoning and examples.	CO1
(b)	Analyze situations where a one-pass assembler would be preferred over a multi-pass assembler, supporting your answer with practical examples.	
(c)	Differentiate system software with other softwares. Explain the various types of system software and their key functions.	
q2	Attempt Any two	
(a)	Differentiate the linkage editor and linking loader. Describe the different data structures that are used to implement the linking loader. Consider the following fragment of a SIC (Simplified Instructional Computer) program:	CO2
	<pre> DATAI START 3000 EOFI BYTE 'C*STOP' TEN WORD 10 INDEX WORD 0 RETURN RESW 1 SIZE RESW 1 ARRAY RESB 1024 BEGIN STL(14) RETURN LOOP2 JSUB(48) INPUTR LDA(00) SIZE COMP(28) INDEX JEQ ENDPRG JSUB(48) OUTPUT J3C) LOOP2 </pre>	
(b)	<pre> ENDPRG LDA(00) EOFI STA(0C) ARRAY LDA(00) TEN STA(0C) SIZE JSUB(48) OUTPUT LD(08) RETURN RSUB(4C) </pre>	
	<pre> DEVICE BYTE 'XFI' LIMIT WORD 1024 INPUTR LDA(04) INDEX LDA(00) INDEX TDE(0) DEVICE JEQ(30) ILOOP RD(D8) DEVICE COMP(28) INDEX JEQ(30) EXIT STCH(54) ARRAY,X </pre>	
Given Memory Layout:		

Memory Address	Contents
3000	53544150 007A6C09 00000000 00000000
3010	XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX
4000	XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX14
4010	30094820 3000000C 77
4020	77 300F0030 77
4030	77 30094C00 77
4040	00300F60 7 4B30B209 7
4050	7 0FXXXXXXXX XXXXXXXX XXXXXXXX

i. Using the Load-and-Go Assembler technique, fill in all the missing (?) memory contents by calculating the address references and machine code values.
ii. Construct the Symbol Table for all symbols defined in the program, showing their addresses and lengths.

(c) Argue on the requirement to generate unique labels during the macro expansion? Write the method that can be used to generate unique labels during macro expansion.

q3 Attempt Any two 20 Marks

Given the following macro definition:

```

END1 MACRO &DEVICE=&F, &BUFSTR, &BUFLN, &ENDIN=&7, &MRL
    CLEAR X
    IF (&MRL NE '')
        -LDS #&MRL
    ELSE
        -LDS #8192
    ENDIF
    $REPEATD =X'DEVICE'
    JEQ $REPEAT
    RD =X'DEVICE'
    &COUNT SET 1
    WHILE (&COUNT LE &ENDIN)
        COMP =X'00000&COUNT'
        JEQ $BREAK
        &COUNT SET &COUNT+2
    ENDW
    STCH &BUFSTR,X
    TIXR S
    .LT $COUNT
    $BREAK STX &BUFLN
MEND

```

(a)

Expand the following macro invocation statements using the above macro definition:

```

i. END1 BUFSTR = B1, BUFLN = LEN, MRL = 512
ii. END1 DEVICE = B8, BUFSTR = BUFARA, ENDIN = 8, BUFLN = 13, MRL = 4096

```

A portion of machine code from a simple hypothetical processor (with fixed 1-byte opcodes and 1-byte operands) is given below:

Memory Address Machine Code (in Hex)

2000	14
2001	05
2002	23
2003	09
2004	30
2005	00
2006	40

(b)

The instruction set of the processor is as follows:

Opcode (Hex)	Mnemonic	Description
10	LOAD A, addr	Load accumulator from memory
14	LOAD A, const	Load accumulator with immediate constant

- 20 ADD B, addr Add memory operand to register B
23 ADD A, const Add immediate constant to accumulator
30 STORE A, addr Store accumulator to memory
40 HLT Stop execution

- i. Disassemble the given machine code into its equivalent assembly language instructions.
ii. Show the corresponding address and mnemonic for each instruction.

Consider the following SIC/XE source program (CO3)

0000	END3A	START	0	
	EXTDEF	LABS1, LABEL		
	EXTREF	LABS2, LABS3, LABEL2, LABEL3		
0020	LAB1	LDA(00)	LABS1	
0023	LAB2	+LDT(74)	LABS2+4	
0027	LAB3	LDX(04)	#LABE1-LABS1	
0040	LABS1	EQU	*	
0054	LABEL	EQU	*	
0060	LAB4	WORD	LABS2-LABS1	
	END	LAB1		

0000	END3B	START	0	
	EXTDEF	LABS2, LABEL2		
	EXTREF	LABS1, LABS3, LABEL, LABEL3		
0036	LAB1	LDA(00)	LABS1	
003A	LAB2	+LDT(74)	LABS2+4	
003D	LAB3	LDX(04)	#LABE1-LABS1	
0060	LABS1	EQU	*	
0070	LABEL	EQU	*	
	END			

0000	END3C	START	0	
	EXTDEF	LABS2, LABEL2		
	EXTREF	LABS1, LABS3, LABEL, LABEL3		
0018	LAB1	LDA(00)	LABS1	
001C	LAB2	+LDT(74)	LABS2+4	
0020	LAB3	LDX(04)	#LABE1-LABS1	
0030	LABS1	EQU	*	
0042	LABEL	EQU	*	
	END			

Based on the above program complete the object program on control section

HEND3A (1)
D (2)

ROZLABS203LABS304LABE205LABE3

T (3)

T00006003FFFC0
M000024 (4)
M000054 (5)
E000020

HEND3B 00000000007F
D (6)
R (7)

T0000360B0310000077202705100000

M000037 (8)
M00003E (9)

E
HEND3C 0000000000051

M000019 (10)

E

q4 Attempt Any two

20 Marks

CO4

Explain any two with suitable examples

- (a) i. Linker vs Loader ii. Macro parameter concatenation iii. Keyword macro parameters

Consider the following code segment

```

HMAIN200000000120C
...
T0000200C1420334400002C3566770034
...
M00002405

```

(b)

- i. Explain how the process of program relocation is affected by the use of a modification record in the object program.
ii. If the program is loaded starting at address 2B10, compute the final values stored at memory locations 2B32 and 2B33 after relocation is performed.

- iii. Describe the efficient technique used in the SIC (Simplified Instructional Computer) for producing a relocatable object program, and explain how the loader handles relocation during execution.

Explain the significance of dynamic linking in modern computer systems with an appropriate example.

Describe various steps performed by the loader during the dynamic linking process.

(c)

20 Marks

CO5/CO6

q5 Attempt Any two
Write the LEX/YACC program to convert a decimal number into binary number. Write the various commands that are required to run the written program.

Demonstrate that the following grammar is ambiguous by considering the string:

if c1 then if c2 then s1 else s2

Given Grammar:

<statement> → if <cond> then <statement>
| if <cond> then <statement> else <statement>
| s1 | s2

<cond> → c1 | c2

(c) What are debugging systems? Explain in detail the main functions, features, and capabilities of a debugging system used in program development and error detection.

